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Earthmoving for the extraterrestrial

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Abstract

The United States spaceport, and more importantly, its technical landscape, operate in the background for the technological and political progress in pursuit of the extraterrestrial. Throughout the construction of the launch complexes on the Florida coastline, earthmoving became standard practice to elevate rocket pads above sea level and protect against rocket blasts. However, a more extended history of earthmoving at Cape Canaveral is necessary. From Earth's early geological formations and indigenous burial mounds to the modernization of rockets, Cape Canaveral presents itself as an evolution of terrestrial form. And still, earthmoving continues today. In 2016, an article advised the greatest threats to NASA's landscape are rising sea levels and hurricanes, causing substantial erosion to the beach, leaving active and historically significant launch facilities at risk. Cape Canaveral beaches are now preparing for additional dredging, importing new soil, and raising beach and dune elevations. This article reveals a critical history of place-based science on Cape Canaveral through an evolution of earthmoving practices, from cultural commemoration, extraterrestrial imagination, and contemporary environmental crises.

Keywords

Infrastructure, NASA landscapes, technical lands

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In 1950, the United States Air Force selected the Central Florida eastern coastline to construct its missile operations and rocket launch facilities. Cape Canaveral proved to be a "suitable" decision for military security, proximity to the equator for rocket efficiency entering orbital trajectory, and safety as failed rockets could discharge in the ocean.

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The spaceport, and more importantly, its technical landscape dominated by earthmoving practices, operate in the background for the technological and political progress in pursuit of the extraterrestrial. However, the context of earthmoving requires a much broader history, including Cape Canaveral's early natural, cultural, and architectural formations. Departing Earth requires the moving of it. The mechanical landscape of the launch complex displaces more than just soil; it erases natural and cultural histories. Historical tensions between the rise of technology and environment have been widely discussed and negated in social theory and the design fields. Instead, in the context for departing Earth, this dichotomy provides an even deeper thread between land itself and the ways we consider outer space. This essay sugggests the changes of landscape over time, and *how* land is formed, is a condition entangled with the affordance of the US launch complex and Cold War infrastructural landscape.

The transcendence from a terrestrial surface to extraterrestrial space is made by the fragile, and temporal, infrastructures forming techno-spatial landscapes on Earth. And here, in this history of space exploration, technological landscapes generate a continuous gateway of opportunity to see the universe and simultaneously ourselves. It has recently been argued that "environing and technology" offer influence on both the material and immaterial world, including the "tweaked natural geosystemic processes" (Sörlin and Wormbs, 2018). This suggests that reading the Anthropocene includes historical landscapes and imaginaries about the world. Just as we see lands shaping and being shaped by geological formations *and* cultural imagination, so too does fanatically celebrated history of militarized landscape for the extraterrestrial inform an ideology of technology, society, and place (Daniels, 2017). Imagination is as valuable as hard science (Ferguson, 1992). For architects and engineers, the leveling of land becomes fundamental in how we position ourselves in the environment—an attempt to minimize interruptions on the working surface of the earth, and in the spaceport case specifically, driven by an extraterrestrial imaginary. Transformations of the technical landscape is constantly shaping and reshaping an image for constructing military power—images of the cultural landscape itself and the icons used for informing historical claims. Earthmoving plays a key role in the organizational tools for production and operations in the historical ports of power. However, the public image supporting extraterrestrial pursuit rarely finds its way back to the land itself (Berry, 2012; Klinger, 2019).

The coastal configuration of Florida's peninsula did not always exist. Submerged by the early planet's high seas lies its geologic past. Known as the Florida Platform, this large flat land mass fluctuates between exposure and ocean coverage. During the Miocene epoch, around 23 to five million years ago, sea level increased due to a warming planet. Throughout this period, powerful ocean currents moved through Central Florida depositing phosphate and materializing into a collection of accumulated minerals (Hine et al., 2017) (Figure 1).

Giant sheets of ice formed during the glacial period, causing sea levels to drop more than 100 feet compared to Central Florida's elevation today. The exposed Florida Platform during the last ice age was dry and consumed with radically cold winds. This wind pushed sediment and sand into U-shaped mounds identified by draping "arms" known as parabolic dunes. By the Holocene period, the climate began to warm, melting away the ice. As sea level rose, the dunes held their shape and became undulating hills, now able to support vegetation and early growth of sea oats as humidity increased. This natural formation and rise in sea level caused freshwater hidden beneath the surface to percolate through aquifers creating Florida's famous springs and mixture of brackish waters along the coast (Rodrigues et al., 2022).

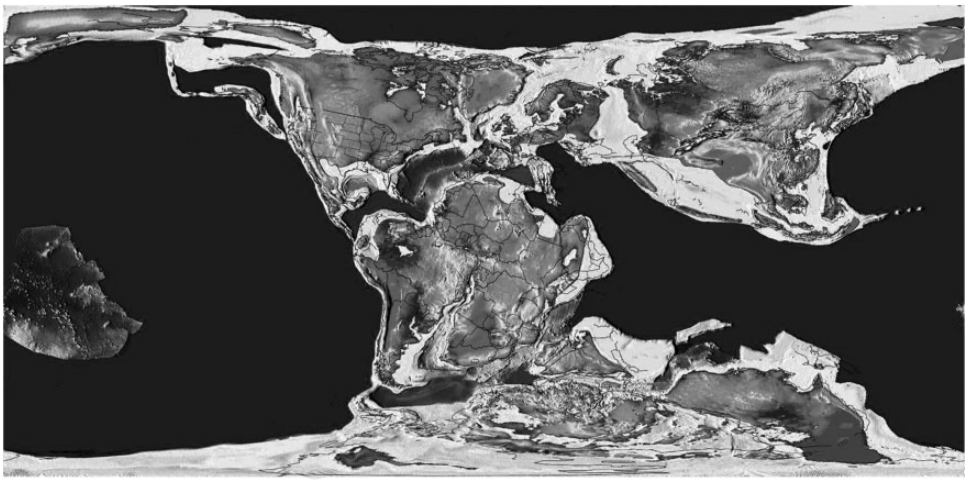

Figure 1. Paleogeographic map. Note: present-day Florida underwater, Paleo-digital elevation models for the Phanaerozoic (Source: Christopher R. Scotese and Nicky Wright).

Specifically, the eastern coast of Central Florida waters continued to rise and fall during the end of the Pleistocene and beginning of the Holocene. Land instability led to the creation of a Beach Ridge adjacent to the modern-day Cape Canaveral coast. Over a dozen sets of beach ridges are made through varying deposits spanning the last 5700 years. These early episodes of earthmoving, leading to the formation of beach ridges, reveal the fluctuating relationship between water and land at the Florida Platform (Rodrigues et al., 2022).

In this celebrated region of Central Florida, long before any rocket launches, indigenous people already settled in the area, including Miccosukee, Mascogo, Seminole, and Ais. Over 1000 years ago, particularly the traveling Ais people inhabited Cape Canaveral wetlands (North Brevard Business Directory, 1967). Because hunting and fishing were key to their survival, the Ais people moved in small groups up and down the rivers and coastlines of southern and Central Florida. The native community relied on wildlife, and its society was heavily influenced by the abundance of food from the marshy and extensive waterways. Many of the villages lived along the beach areas formed by the undulating parabolic dunes. Fishing and hunting were particularly abundant along the brackish waterways, lagoons, and specifically sourced through the Indian River. But with the rise of waters from the previous geologic period, the arrival of mosquitos populated the marshy waters causing the Ais communities to move into higher elevations during the summer months.

Subtle manipulations of dune formations, waterways, and winds altered the land and the regional flora and fauna. Yet, the Ais people, as one of the many indigenous groups living in the area, were adaptable and resilient. Although mobile across the undulant wetlands, Ais people created commemorative burial mounds for their lost ones. Land is sculpted to mark the dead. Some burial mounds are known to have reached up to 25 feet high and stretch over 100 feet across (North Brevard Business Directory, 1967). Modern archeologists from the Florida Public Archaeology Network and archaeologist for the Seminole Tribe of Florida note that at least over three dozen mounds have been found across this region including the present-day Cape Canaveral landscape (Livingston, 2022) (Figure 2).

Even the geographical naming of the region is curiously linked to land formation and colonization. The late 16th century Spanish explorers were the first to name the region *Cabo*

Figure 2. Fort Pierce Midden Mound, Brevard County, FL (Source: Florida Museum of Natural History/ Clara Miller).

Cañaveral—a Spanish term meaning "cape reedbed" or "cape sugarcane plantation." According to the National Geographic, a "cape" is defined by the high point of land that extends the furthest into a body of water. But this land formation as a "cape" occurs through the long history of sediment formation, as discussed previously, but also through other earthmoving practices. Once Spanish explorers began to migrate and colonize the area in the 16th century, new tools from Europe colonized the Ais region—inevitably erasing much of Ais people, culture, and integrity of their communities. As described by anthropologist Sarah E. Vaughn, "industrialism is naturalized as Indigenous to the landscape, and Indigenous bodies and knowledge are rendered expendable, marginal, and pollutable" (Vaughn et al., 2021). This suggests that in this context, the mechanization of the US launch infrastructure dismissed the nature and cultural history of the region. In other words, the displacement of land is the displacement of indigenous cultural history.

European military and outside political authority swept over the land. With the British asserting control from 1763 to 1784, and the Spanish again until 1821, the United States eventually established the territory of Florida (Adams, 1997). Stricken by the conflicts between the United States and the Seminoles known as the Seminole Wars, the United States Civil War, and territorial claims over its geographical location—including the militarization of the land itself—the history of the Cape Canaveral region consistently underwent changes of power and spatial structure.

The term wilderness has been problematized by several scholars. Specifically valuable here in this context, wilderness is to be understood as an invented idea from an American obsession (Cronon, 1995; Guha, 1989)—a constructed imaginary to enable the designations of land for other uses. In 1876, a series of "house of refuge" stations were constructed by the

United States Life-Saving Service, the agency preceding the U.S. Coast Guard. These unmanned life-saving stations were sited along the Florida Atlantic coastline every 20 miles. The modest rescue shelters were stored with food and supplies for ship-wrecked sailors. But as the refuge sites were little used, many were abandoned or repurposed. In 1889, just 13 years after their initial construction, wealthy ornithologist Charles Barney Cory from Massachusetts reclaimed one of the "house of refuge" sites. With Harvard elites from Boston, Cory established the "Canaveral Club," a luxurious hunting and fishing camp (Marconi, 2010). Transformed from an empty shelter on the wetland coast, the region was considered, by Cory and his friends, a vacation in the southern wilderness at the edge of the nation. Supposedly located in an isolated and remote wilderness as marketed by the Cory group, the former "house of refuge" site was treated as luxurious grounds of an astonishing 18,000 acres in the 'wild.' Quickly after the recreational explorers purchased the area, club members quickly began altering the landscape and ultimately its wildlife (McQuade, 2016). The group became the first to dredge the lagoon systems in the region, allowing their gasoline-powered engine (10 horsepower) sternwheelers to enter the Banana Creek and access acreage along the Indian River more easily. This altering of land made it possible for an increase in boat transit—including shallow draft steamboats and private yachts—and helped support an increase in tourism. The Canaveral Club became wildly popular for outdoor enthusiasts and tourist's wealthy enough to afford the $5000 member fee in 1900 (approximately $150,000 in 2022 dollars). After Cory's death in 1921, the land went through several changes of ownership. For example, segments of the site were occupied by a local Boy Scout troop camp site. Ultimately, due to overexploitation of territory, early dredging, clearing of lands, and slaughtering of local white-tail deer, black bears, water ducks, and alligators for commercial use, the local wildlife was left devastated—including a damaged ecology of fluctuating land and water cycles. But earthmoving alterations to this wetland continued (Figure 3).

The earliest of rocket launch sites required the design of earthmoving techniques. The launch pads and facilities at Peenemunde had been under the architectural direction and coordination of Albert Speers, the well-known German architect who worked on many military complexes during the Second World War. A dense series of mechanical conduit lines organized the complex's testing facilities. Such lines demonstrate an early example of the application of underground networks and technology management, as well as a merger of systematic operations and ground reconfiguration. Bounded by an "embankment" of built-up earthwork, the launch pad at Peenemunde and its electrically networked systems became the model for Americans at Cape Canaveral. Influence came from two German pioneers in rocket development, Wernher von Braun and Dr. Kurt Debus. Long before NASA's establishment, the renowned von Braun, however, had already been recruited from the experimental test stands at Peenemunde. Working for United States Army Intelligence starting in 1945, von Braun moved to Fort Bliss in Texas, then New Mexico, and later transferred to Huntsville, Alabama, as the director of the Army Ballistic Missile Agency's Development Operations Division. Eventually, von Braun worked with NASA on Apollo's Saturn rocket development. Associated with the first German test stand site as early as 1942, von Braun and Debus were likely significant in shaping designs for the new military launch infrastructure in the United States. Spaceport infrastructure became a systematization of military land "designed" by *and* appropriated from German experts working previously in the earth-walled German Peenemunde launch complex.

By 1949, the United States purchased a significant portion of Brevard County land for the U.S. Naval Air Station, which included the Canaveral Club grounds. Quickly upon acquisition, the Air Force burned down the remaining structures of the luxury hunting

Figure 3. C.S. Robinson Aerial Surveys (1943), U.S. Department of Agriculture (Source: Brevard County Library Archives).

camp. And shortly after, the Air Force launched its first V-2 German-designed rocket in July 1950. Under the new name, Cape Canaveral Air Force Missile Test Annex, the site rapidly increased its launch complex infrastructure. Much of this effort was in direct support for the testing of ballistic missiles. However, the architects and engineers had little precedent for designing such military infrastructure that included the necessary spaceport anatomy—a semi-buried concrete blockhouse, elevated launch pad, and rocket service tower—especially in a time of political expediency formed by the rise of the Cold War. Eventually, an astonishing 70% of all launch complexes at Cape Canaveral were constructed over a brief seven-year span ranging from 1950 to 1961. By the time NASA was established in 1958, 23 of the total 36 launch complexes were already in place. During the construction of launch site blockhouses and rocket pads, earthmoving became standard practice for protection of the

Figure 4. Launch Complex 17 Blockhouse (1956, initial construction), Cape Canaveral Missile Test Annex, Brevard County, FL (Source: Historic American Engineering Record/Library of Congress).

blockhouse against the rocket blast and to elevate rocket pads above sea level. The rise of launch complex and space race archaeologies may signal cultural memory and provide an index of technical objects for extraterrestrial and interplanetary pursuits (Alonso, 2016; Capelotti, 2010; Gorman, 2003) (Figure 4). But what kind of landscape was the military constructing on? What can we learn between the relation of the celebrated objects and technical lands for departing earth?

The aerial photographs from the U.S. Department of Agriculture in 1943 reveal a completely different landscape compared to the militarized sites constructed in the 1950s. A series of wide sweeping dunes stretched across the Cape. This elaborate network of sand dunes is naturally formed by the interaction of air and water. Constantly in flux, these dunes helped alleviate and protect land against storms from the Atlantic Ocean. As noted by the National Parks Service, "At times the potential fury of the ocean breaks loose and the shoreline is lashed by northeastern storms of hurricanes . . . Canaveral's 20 foot high dunes are a barrier, a dike, that prevents the flooding of inland areas," a barrier for protection and disaster (National Parks Service, 2020). Perhaps the most vital feature of the dunes is its ability to move as "the sand gradually accumulates, reshaping and expanding the dune" from a gradual circulation of winds and waves.

In 1962, NASA selected the exact location of the "Canaveral Club" for its highly celebrated Apollo launch complex pads 39A and 39B (Daniels, 2017; Westwood et al., 2017). Buildings were not of most importance, and instead the acquisition of land was vital, including the disposition of citrus groves along adjacent property. Even though resistance from the growers occurred, land was eventually purchased by NASA and the agency began

quickly shaping the earth. By dredging hydraulic fill from the Banana River and pumping it into the construction site, engineers elevated the rocket launch pads to sit above the shallow sea level. NASA, under the guidance of the Corps of Engineers, initiated massive wharf construction with a series of barge canals and turning basins. Two large pyramid-like *sandpiles* were constructed to elevate the launch pads used to power the Saturn V rocket to the moon (Benson and Faherty, 1978). In fact, the technology used to support the new mobile launch complex came from common earthmoving practices. The steam-shovel crawler was used for raw extraction and surface mining purposes, primarily for coal in Kentucky. The mining equipment was adopted to construct the Missile Crawler Transporter Facility and used as a vehicle for transporting the rocket from the Vertical Assembly Building to the elevated launch pad. This is a clear indication of NASA's use of both high and low technology to quickly assemble the spaceport under political pressure during the Cold War context (Figures 5 to 7).

The crawler transporter and rocket travelled along a paved three-mile-long crawler way. To make this possible, NASA engineered large subterranean water channels known as culverts for the crawler way to traverse the Banana Creek. The so-called "remoteness" of the rescue stations claimed by United States Life-Saving Service, the Canaveral Club, and NASA's interests to use the Merritt Island Wildlife Refuge to form a safety buffer is greatly transformed into a landscape of machines and objects for the departure of earth. It turns out the crawler way has created severe hydrological problems with the lagoon systems today. Trade winds here come from the Southeast and generate a cyclical pattern of water flow across the Banana River, into Mosquito Lagoon, and down through the Indian River (National Parks Service, 2020). Much of this is caused by a blockage at the intersection of the crawler way and Banana Creek. As described above, this complex system of water, wind, and sand allows for the parabolic dunes to continue gradual sediment accumulation. Apollo's crawler way constructed by engineered infrastructure of earthmoving and blocking of channels across the rivers and lagoons disrupts and alters the environmental water flow and dune systems (Figure 8).

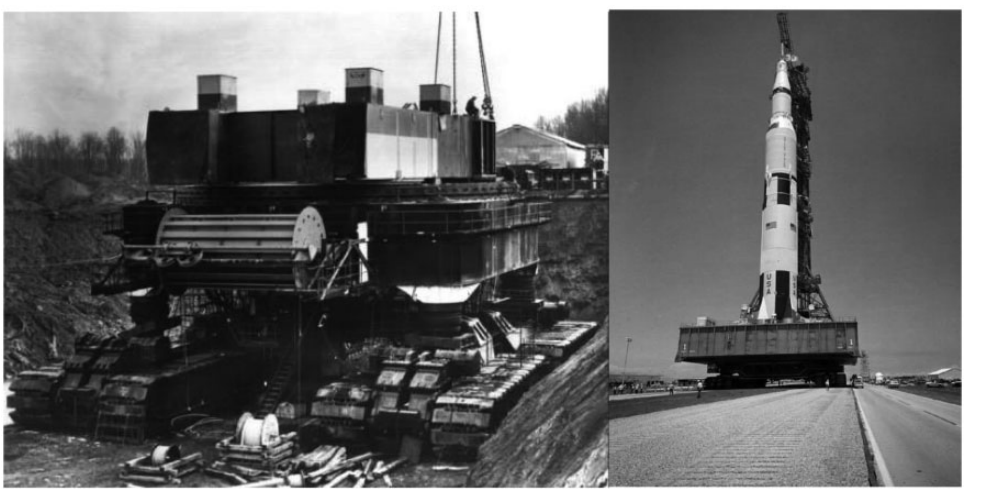

Figure 5. Left: Steam-shovel Crawler (date unknown), Kentucky, USA (Source: NASA Archives/Benson and Faherty). Right: Apollo, Saturn V Rocket on Missile Crawler Transporter (1968), Kennedy Space Center, Merritt Island, FL (Source: NASA Archives).

Figure 6. Left: C.S. Robinson Aerial Surveys, Banana Creek with first dredging (1943) (Source: U.S. Department of Agriculture)/Right: Aerial Surveys, Launch Operations Center (1969), Crawler way, Vehicle Assembly Building, and Launch Control Center (Source: Brevard County Library Archives).

Figure 7. Launch Complex 39A Pad construction (1963), Kennedy Space Center (Launch Operations Center), Merritt Island, FL (Source: NASA Archives).

Figure 8. Launch Complex 39 Crawlerway construction (1963), Kennedy Space Center (Launch Operations Center), Merritt Island, FL (Source: NASA Archives).

Earthmoving for reaching the extraterrestrial from Cape Canaveral concerns three landform typologies: burial mounds (and a shell midden mound), launch complex infrastructure, and a history of dredging—a link between culture, landscape, and outer space (Redfield, 2000). All three have artificially manufactured ground to block and manipulate water courses and natural wetland ecologies. Yet, each episode is produced by radically different cultural, political, and social contexts—much like the troubled boundaries between systems and the extreme environment in the work of Valerie Olson's work where the technical and political environment merge (Olson, 2018). Not only do time periods distinguish their differences, but also intention, imagination, and future unpredictability. The modern rationalized techno-spatial landscapes of scientific and environmental struggle illustrate an Indigenous culture and varied commemoration in the shadows of modern earthmoving alterations. Early Cold War anxiety over nuclear conflict generated an unusually rapid speed of proving grounds, a site designated for military tests in science and technology (Ponte, 2003; Woodward, 2004). As development of one missile technology emerged, so too did its associated infrastructure, requiring new landforms to be constructed.

Likewise, the rocket technology required a loose, flexible infrastructural landscape to support changes in space exploration due to political and economic fluctuations. On 28 November 1963, six days after President John F. Kennedy's assassination, successor President Lyndon B. Johnson renamed the space complex *Cape Kennedy*. President Johnson's dedication of "Cape Kennedy" helped secure Kennedy's political legacy and image, both tied to NASA's political pursuits. Just as President Kennedy became associated with national progress and aspirations for reaching the moon, NASA's modern architectural forms in the remote wilderness were just another attempt to, or in the words of cultural historian Martin Jay, "linking the human desire to mold the environment for aesthetic purposes with the domination of nature" (Jay, 2011). Ironically, launch sites have two unique characteristics. First, the requirement for national security physically bounded the

territory of the spaceport, restricting accessibility while, paradoxically, rocket activities greatly increased our access to the extraterrestrial by placing probes, satellites, and humans in space. Second, next to Cape Canaveral we find a landscape protected ecologically by its adjacent technology. Merritt Island national Wildlife Refuge was established on 28 August 1963, by the US Fish and Wildlife Service, designating the area for ecological conservation and protection of migratory birds. Still today, this highly preserved estuary is considered to be one of the last "natural" beaches on the Florida coast. Yet, the US military had already been launching rockets on this land for over 10 years before its inception, prohibiting any development to occur in the area.

This Merritt Island National Wildlife Refuge is protected by the intense techno-spatial logistics and facilities for launching rockets (due to isolation from any nearby urban development for safety precautions). This historical dichotomy between "nature" and "technology" has a more interconnected relationality within the spaceport complex, as is commonly understood. Yet here the evolution of earthmoving practices offers a particular lens to describe how soils in the landscape support continuous reframing of culture, technological speed, and extraterrestrial imagination. In fact, the militarization and launch complex infrastructure have ironically saved and protected the so-called natural environment. Therefore, the reversal is true, the wildlife conservation is a techno-spatial environment enacted by the imbrications of global Cold War militarization.

The notion of a "spaceport" identifies distinct features in the operations of the historical launch site for departing Earth. The term "spaceport" first appeared in 1930 from the American science fiction magazine titled, Amazing Stories, which began publication in 1926 and running until 1959. As portrayed in the fictional setting, "the space-ports at the three great cities, were, of course, occupied or blockaded by the Tellurian fleets; and Doane was obliged to make his bases of operations the lonely craters that once had been pirate strongholds" (Breuer and Williamson, 1930). Curiously related to the evolution of cities, the term "spaceport" in early science fiction appears as a condition linked to power, tied to a history of former occupation. Moving from sea, to air, to space, the spaceport is a critical landscape of projected imaginary and expanded territory and consequentially generating cultural memory (De Monchaux, 2017; Sammler and Lynch, 2019).

The present-day office seal of Brevard County marks its history as a central connection to exploration with caricature illustrations: a 16th century European colonial ship reaching the Florida coast lined with orange trees and a 20th century Apollo capsule and astronaut landing on the moon. Nothing on the seal describes a history prior to colonial conquest or the modern and highly romanticized aerospace historical episode. It could be argued, therefore, that these forms—the ship and the capsule—are not just symbols celebrating regional or technological triumph, but also indications of colonial powers, spatial control, and the conquest of territory over people and the environment. Perhaps even a celebration of technology *over* nature (Figure 9).

Throughout the construction of the launch complexes, earthmoving became standard practice to elevate rocket pads above sea level and used for protection against the rocket blasts. And today, Brevard County still claims early Ais communities lived on "crude and flimsy" wooden structures (North Brevard Business Directory, 1997). Supporting its claim of one history as wilderness and its inference that the region was inherently unclaimed due to nomadic society. Brevard County's historical account simplifies the Ais' culture and overlooks the benefits of a remote and mobile dune, wind-driven landscape. However, the construction and designation of the so-called "wilderness" as an idea is embedded in a political construction made from assertions of territorial control over land. All of which the state, or

Figure 9. Seal of Brevard County Florida (Source: Brevard County, FL).

local government, attempts to generate specific narratives of historical geography for future interests, economic and political.

Two important designations are made in Central Florida. On one hand, the foundations of colonial pursuit repeatedly designate the region as a crude wilderness to claim land, and on the other, military delimiting of the landscape aligned with the so-called preservation of it. This "trouble with wilderness" (Cronon, 1995) suggests that the preservation of Merritt Island adjacent to rocket launches are ultimately entangled in the same view of cultural politics over land and territory. Just as the Everglades National Park were a culturally constructed idea and a fabricated landscape of wetland preservations, constructing Cape Canaveral's history as a wilderness led to the support of a national image for military-based missile exercises and space exploration during the Cold War. Success of reaching the Moon relied on the notion of "wilderness" as an open landscape and described as having little geologic or cultural pre-modern history.

Perhaps more important in our understanding of the impact of this landscape is not the colonial act required claims of "wilderness," but also that the landforms have histories embedded within them. Even though the native Ais communities were in fact mobile, traversing the wetlands, they are left with traces of cultural memory in the landscape. Although only a few burial mounds of built-up earth along with the Turtle Mound Cape Canaveral National Seashore (a shell midden formed from oysters by the Timucua people) remain. These landforms of ceremonial earthworks indicate a history of cultural presence yet silenced by colonial obsessions; from the arrival of the Conquistadores to landing on the Moon. For the Ais, the wetlands became the cultural foundation for cultivating community and survival, such as their fishing and hunting techniques with the land, while the development and operations of the Eastern Test Range allowed the United States military to incorporate wilderness as a criterion for maintaining, and increasing, its territory of extraterrestrial space and laying groundwork for future imaginations of sovereign power. And it is in the wilderness that strategic earthmoving practices for militarizing land veiled by civil safety—including extended geographies for tracking and monitoring—are enabled (Elden, 2013). Spatial geography of the missile range and configuration of its tracking

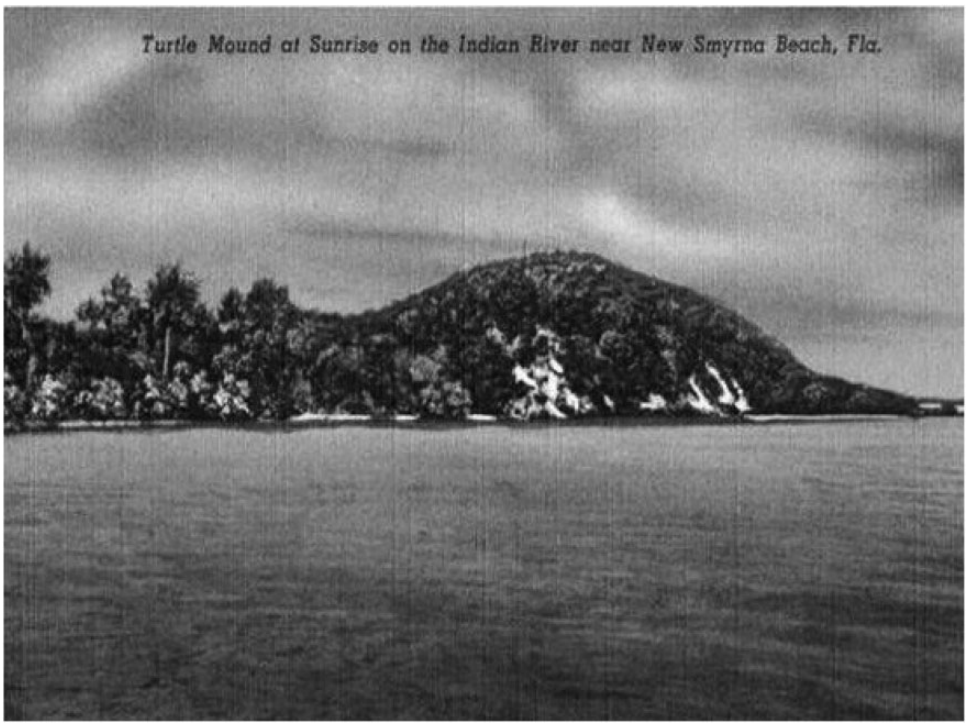

Figure 10. Turtle Mound Post Card (c. 1970s), Canaveral National Seashore, New Smyrna Beach, FL (Source: University of Central Florida/National Parks Service).

stations within *invisible* space are as much a political and spatial strategy as they are a science-based technical solution (Nesbit, 2020; Nesbit and Nichols, 2020) (Figure 10).

Histories of terraformations can reside simultaneously. Today, much of the blockhouses remain abandoned in the Central Florida wetlands. But so too do the existing shell mounds. The American claims over wilderness deterritorializes the launch complex, a condition of militarized landscape infrastructure, and represents the rise of an operationally remote military activity that accelerated well into the Cold War. But as described in the Brevard County seal, it is the temporal-imaginaries and the images of such terraforming history that became suppressed. Moreover, the launch complexes themselves become the surrogate for a radicalization of urbanizing humanity—an American wilderness as a site for technology in the purposely invisible and geopolitical positioning on and off the Earth. All the while, such political and cultural interests for departing Earth require earthmoving practices explicitly leaving a damaged landscape left behind.

Earthmoving persists. In 2016, an article in Gizmodo advised the greatest threats to NASA's landscape are rising sea levels and hurricanes, causing major erosion to the beach leaving active and historically significant launch facilities at risk (Stone, 2016). Cape Canaveral beaches are now dredging, importing new soil, and raising beach and manipulating dune elevations. Recently, the Florida coast was threatened by hurricane Dorian in September 2019. The hurricane generated massive environmental threats to its coastline through a series of storm surges while leaving behind increased erosion along the beach. As the second strongest Atlantic storm in over 70 years, Dorian is a clear reminder of environmental catastrophe up against of landscape of cultural conflict and commemoration, another mode of

Figure 11. Major Beach Erosion, rocket stand in the distance (2017) (Source: NWF Daily News Journal/Jim Tiller).

earthmoving, nonetheless. This is important distinction within the long durée of the region's natural history. The link between the long geological history, indigenous terraforming, and rocket launch infrastructure is related, and matter. The return of climate change impact on the infrastructural landscape is another form of long geological history, yet in this case on the Cape Canaveral beach is a clear example where the cause of landscape deterioration and the earthmoving remedies are interrelated conditions (Figure 11).

The evolution of launch site infrastructure is a way to decipher the evolutions of territory marked by launch site configurations. Initially, it seems the history of earthmoving in the context of the spaceport launch complex captures territories of extraction processes for raw materials, leading to the deployment of vessels, in sea and space, to enable military tactics and political positioning. After a deeper re-reading of the technical land, it seems Cape Canaveral (as a terrestrial surface) continues to be altered and absorb varying degrees of influences time and scale. Eventually, this mode of seeking wilderness as a mechanical landscapes for launching rockets, is well acknowledged. But also, terraforming, and earthmoving practices mark a contingent history and demonstrate how technical lands are formed. Colonial language of remote and distant wilderness is made through the legal claims over land designated by military or scientific land use, and therefore territories are exploited (Galison, 2017; Nesbit and Waldheim, 2022). From the moment of geological resources and the reformation of land, the loss of indigenous culture is eclipsed by settler colonial politics.

Earthmoving practices in the militarized infrastructural launch complex not only leave us with a stratification of ground, fragmented history, and a geopolitical territory leveled and re-leveled. It is now the site of threat. According to the Space Coast Transportation Planning Organization 2018 report on "Sea Level Rise: Vulnerability Assessment," the largest area in Central Florida to be inundated with rising sea is both the Merritt Island

Figure 12. Future of Kennedy Space Center and Cape Canaveral, rendering by author (2022) (Source: Author/#Midjourney).

National Refuge along with Kennedy Space Center and Cape Canaveral Air Force Station. From earthmoving to technical lands, the planetary ocean is now taking back the land that it once covered millions of years ago (Figure 12).

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Biography

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